giving

$$\begin{aligned}\mathcal{L}_{\mathrm{LD}}\phi &\leq -2\gamma l\phi + 2\gamma l H^{l-1}U(\boldsymbol{q}) + \gamma l(l+\mathrm{N_c}-1)H^{l-1},\\ &\leq -\gamma l\phi + \gamma c + \gamma d,\\ &\leq -\alpha\phi + \delta,\end{aligned}$$

satisfying Assumption 2.

The purpose of using the l exponent in the Lyapunov function is to ensure that we can apply the theorem to a wide enough class of observables, as (6.48) is only valid for functions f with magnitude less than or equal to the Lyapunov function. As we have $\phi \geq U^l_{\min}$ and growing like $\|\boldsymbol{p}\|^{2l}$, we can choose l to be sufficiently large to bound a wide class of observables. However, for sufficiently fast-growing functions (such as exponentals) a different Lyapunov function would be required (if their averages do, in fact, converge).

We note that the condition of Assumption 1(ii) can be shown by demonstrating that *Hörmander's condition* holds over the entire space $\mathcal{D}$. This is a condition on the commutators or Lie brackets of vector fields.

The *Lie bracket* of a pair of vector fields $\boldsymbol{u}(\boldsymbol{z})$, $\boldsymbol{v}(\boldsymbol{z})$ is another vector field

$$[\boldsymbol{u},\boldsymbol{v}] = \boldsymbol{v}'\boldsymbol{u} - \boldsymbol{u}'\boldsymbol{v},$$

where $\boldsymbol{u}'$ is the Jacobian matrix of the vector field $\boldsymbol{u}$. In coordinates, if we let $\boldsymbol{w} = [\boldsymbol{u},\boldsymbol{v}]$, then the coordinates of $\boldsymbol{w}$ are

$$w_i = \sum_{j=1}^{m} \frac{\partial v_i}{\partial z_j}u_j - \frac{\partial u_i}{\partial z_j}v_j.$$

The Lie bracket is a bilinear and skew-symmetric operation:

- $[\alpha\boldsymbol{u} + \beta\boldsymbol{v},\boldsymbol{w}] = \alpha[\boldsymbol{u},\boldsymbol{w}] + \beta[\boldsymbol{v},\boldsymbol{w}]$,
- $[\boldsymbol{u},\boldsymbol{v}] = -[\boldsymbol{v},\boldsymbol{u}]$ and $[\boldsymbol{u},\boldsymbol{u}] = \boldsymbol{0}$.

Definition 6.1 (Hörmander's Condition) Let $\mathcal{U} \subset \mathbb{R}^m$ be open. The SDE (6.43) with C^∞ vector fields $\boldsymbol{b}_0,\boldsymbol{b}_1,\boldsymbol{b}_2,\ldots,\boldsymbol{b}_r : \mathcal{U} \to \mathbb{R}^m$ are said to satisfy Hörmander's condition at $\boldsymbol{z} \in \mathcal{U}$ if the vector space spanned by the collection of iterated Lie brackets

$$\boldsymbol{b}_0,\boldsymbol{b}_1\ldots,\boldsymbol{b}_r,[\boldsymbol{b}_0,\boldsymbol{b}_1],[\boldsymbol{b}_0,\boldsymbol{b}_2],\ldots,[\boldsymbol{b}_{r-1},\boldsymbol{b}_r],[\boldsymbol{b}_0,[\boldsymbol{b}_0,\boldsymbol{b}_1]]\ldots$$

(evaluated at $\boldsymbol{z}$) is $\mathbb{R}^m$.

Computing the iterated Lie brackets becomes, in many cases, an arduous task.